

Proposition: Suppose that $x_1, x_2, \dots, x_n$ are identically and independently distributed (i.i.d.) with $E(x_i) = \mu$ and $V(x_i) = \sigma^2$. Then $E(\bar{x}) = \mu$ and $\text{var}(\bar{x}) = \frac{\sigma^2}{n}$, where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

Proof: $E(\bar{x}) = \frac{1}{n} \sum_{i=1}^n E(x_i) = \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n} (n\mu) = \mu$.

$$\text{var}(\bar{x}) = V\left(\frac{1}{n} \sum_{i=1}^n x_i\right)$$

$$= \frac{1}{n^2} V\left(\sum_{i=1}^n x_i\right)$$

$$= \frac{1}{n^2} \sum_{i=1}^n V(x_i)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n^2} \cdot n\sigma^2 = \left(\frac{\sigma^2}{n}\right)$$

$\Rightarrow$ standard deviation of $\bar{x} = \frac{\sigma}{\sqrt{n}}$

Central Limit Theorem: Let $x_1, x_2, \dots$ be a sequence of identically, independently distributed random variables, each with mean μ and variance σ^2 .

Then the distribution of

$$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^n x_i - n\mu}{\sigma\sqrt{n}} \quad \text{tends to the}$$

standard normal distribution $N(0, 1)$ as $n \rightarrow \infty$

Ex: The lifetime of a special type of battery is a random variable with mean 40 hours and standard deviation 20 hours. A battery is used until it fails, at which point it is replaced by a new one. Assuming a stockpile of 25 such batteries, the lifetime of which are independent, approximate the probability that over 1100 hours of use can be obtained. Given that

$$\Phi(1) = P(Z \leq 1) = 0.8413$$

Sol: Let X_i denote the lifetime of the i th battery to be put in use, then find $p = P\{X_1 + X_2 + \dots + X_{25} > 1100\}$

$$p = \underset{\downarrow}{P} \left\{ \frac{X_1 + X_2 + \dots + X_{25} - 25 \times 40}{20\sqrt{25}} > \frac{1100 - 25 \times 40}{20\sqrt{25}} \right\}$$

$$= P\left\{Z > \frac{1100 - 1000}{20 \times 5}\right\}, \left(\because \frac{\sum X_i - n\mu}{\sigma\sqrt{n}} \sim N(0,1) \text{ as } n \rightarrow \infty \text{ by CLT}\right)$$

$$= P\{Z > 1\}$$

$$= 1 - P(Z \leq 1)$$

$$= 1 - \Phi(1)$$

$$= 1 - 0.8413$$

$$= 0.1587.$$

6. Lognormal distribution

Let $Y \sim N(\mu, \sigma^2)$, then $\boxed{X = e^Y}$ has lognormal distribution

$$\Rightarrow Y = \log X$$

$$f_X(x) = \frac{1}{\sigma x \sqrt{2\pi}} e^{-\frac{1}{2\sigma^2} (\log x - \mu)^2}, \quad x > 0, \mu \in \mathbb{R}, \sigma > 0$$

pdf of lognormal distributed RV X .

$$E(e^{tY}) = e^{\mu t + \frac{1}{2} \sigma^2 t^2}$$

$$Y \sim N(\mu, \sigma^2)$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2} \left(\frac{y - \mu}{\sigma}\right)^2}$$

$$\boxed{f_X(x) = f_Y(y) \left| \frac{dy}{dx} \right|}$$

$$\therefore f_X(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2} \left(\frac{y - \mu}{\sigma}\right)^2} \cdot \frac{1}{x}$$

$$= \frac{1}{\sqrt{2\pi} x \sigma} e^{-\frac{1}{2} \left(\frac{\log x - \mu}{\sigma}\right)^2}, \quad x > 0$$

$$Y = \log X \Rightarrow \underline{y = \log x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{x}$$

$$E(X) = E(e^Y) = \underline{e^{\mu + \frac{1}{2} \sigma^2}}$$

$$E(X^2) = E(e^{2Y}) = e^{2\mu + 2\sigma^2}$$

$$\therefore V(X) = E(X^2) - \{E(X)\}^2$$

$$= e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2}$$

$$\underline{V(X) = e^{2\mu + \sigma^2} \{e^{\sigma^2} - 1\}}$$

Ex: Suppose the demand X of a certain item follows a log-normal distribution with mean 7.43 and variance 0.56. Find $P(X > 8)$.

Soln: $e^{\mu + \frac{\sigma^2}{2}} = 7.43$

$$\mu = 2.0, \quad \sigma = 0.10$$

$$\begin{aligned} P(X > 8) &= P(\log_e X > \log_e 8) \\ &= P\left(\frac{\log_e X - 2}{0.1} > \frac{\log_e 8 - 2}{0.1}\right) \\ &= P(Z > 0.79) \\ &= 1 - P(Z \leq 0.79) \\ &= 1 - \Phi(0.79) \end{aligned}$$